人工智能与自动化学院

模式识别与机器学习



第十一讲 卷积神经网络(*Convolutional Neural Networks*)

11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

11.4 应用示例(*Case Study*)

11.1 背景

深度学习存在的挑战和关键技术

- 如何设计或确定网络结构
  - “领域知识”有助于选择网络结构，如计算机视觉中常常用卷积神经网络
- 模型复杂度高导致过拟合
  - “大数据”有助于增加训练集，提升泛化能力
  - “正则化”—如Dropout等手段，增强对噪声的容忍度
- 模型复杂难以找到最优解
  - 仔细选择初始值，避免落入局部极值，包括预训练等
  - 防止梯度消失，采用合适的激活函数、设计更好的网络结构
- 计算复杂度高
  - 平行化、批量化、GPU等

11.1 背景

深度学习存在的挑战和关键技术



九十年代神经网络研究陷入低潮期

11.1 背景

2006年神经网络重归

materials are identical for all configurations. The blue bars in Fig. 1 summarize the measured SHG signals. For excitation of the *LC* resonance in Fig. 1A (horizontal incident polarization), we find an SHG signal that is 500 times above the noise level. As expected for SHG, this signal closely scales with the square of the incident power (Fig. 2A). The polarization of the SHG emission is nearly vertical (Fig. 2B). The small angle with respect to the vertical is due to deviations from perfect mirror symmetry of the SRRs (see electron micrographs in Fig. 1). Small detuning of the *LC* resonance toward smaller wavelength (i.e., to 1.3-μm wavelength) reduces the SHG signal strength from 100% to 20%. For excitation of the Mie resonance with vertical incident polarization in Fig. 1D, we find a small signal just above the noise level. For excitation of the Mie resonance with horizontal incident polarization in Fig. 1C, a small but significant SHG emission is found, which is again po-

Reducing the Dimensionality of Data with Neural Networks

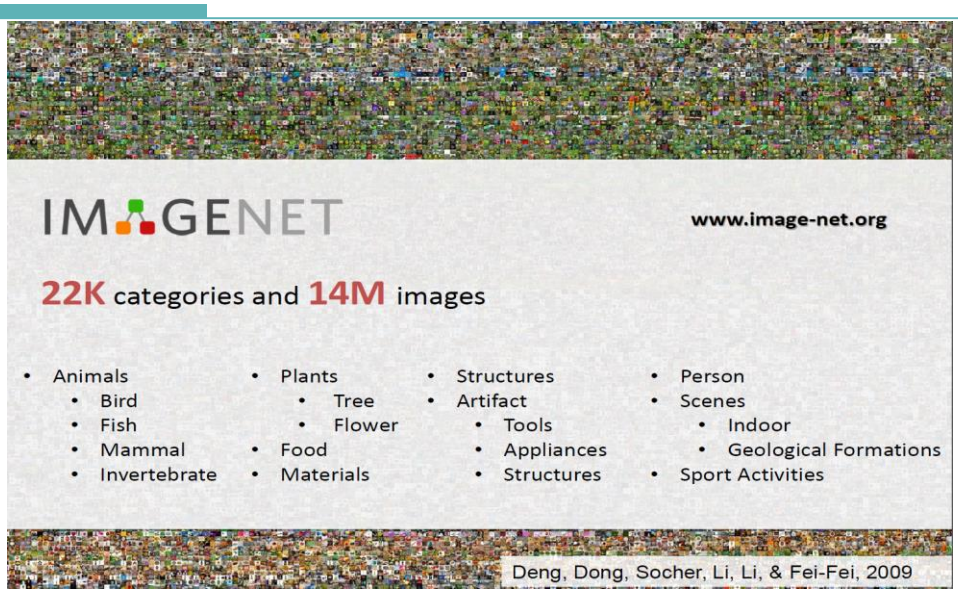
G. E. Hinton* and R. R. Salakhutdinov

High-dimensional data can be converted to low-dimensional codes by training a multilayer neural network with a small central layer to reconstruct high-dimensional input vectors. Gradient descent can be used for fine-tuning the weights in such "autoencoder" networks, but this works well only if the initial weights are close to a good solution. We describe an effective way of initializing the weights that allows deep autoencoder networks to learn low-dimensional codes that work much better than principal components analysis as a tool to reduce the dimensionality of data.

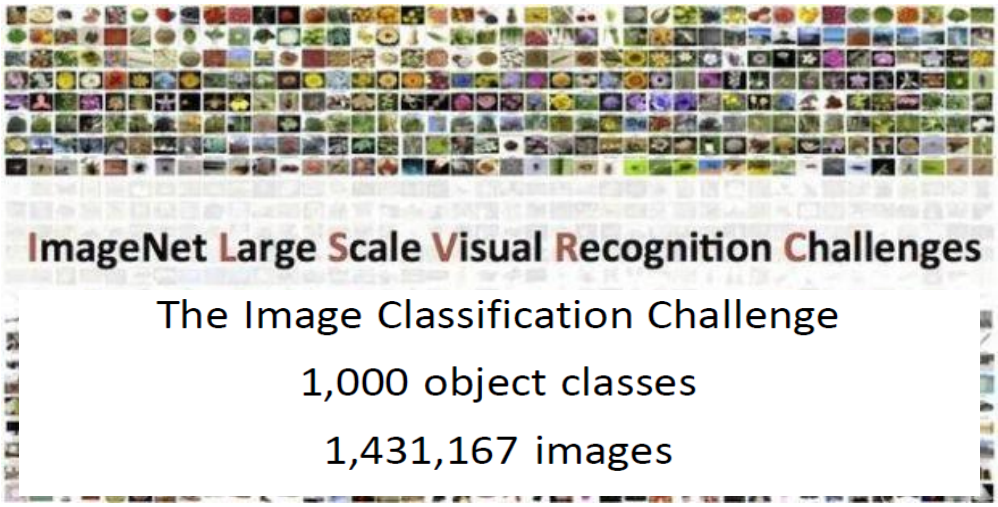
Dimensionality reduction facilitates the classification, visualization, communication, and storage of high-dimensional data. A simple and widely used method is principal components analysis (PCA), which finds the directions of greatest variance in the data set and represents each data point by its coordinates along each of these directions. We describe a nonlinear generalization of PCA that uses an adaptive, multilayer "encoder" network

504 28 JULY 2006 VOL 313 SCIENCE www.sciencemag.org

11.1 背景

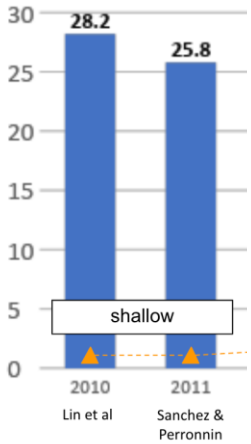


11.1 背景



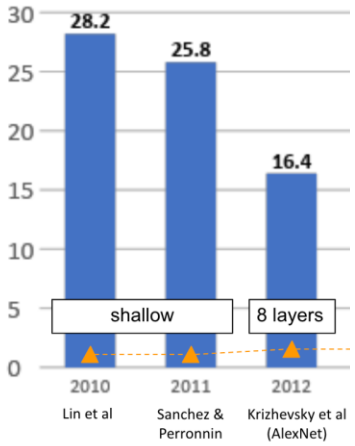
11.1 背景

历年来 ImageNet 挑战赛性能(错误率)排名



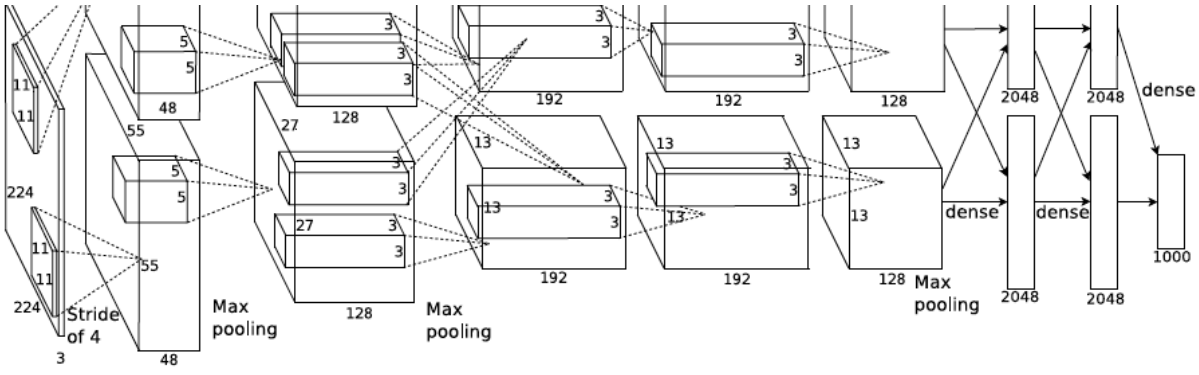
11.1 背景

历年来 ImageNet 挑战赛性能(错误率)排名



11.1 背景

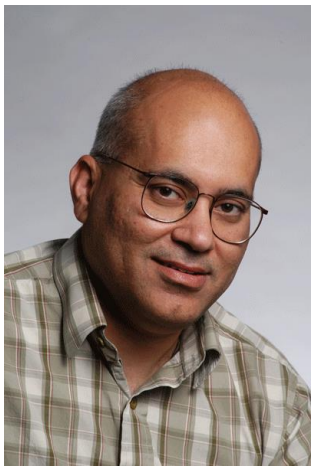
AlexNet in ImageNet 2012



11.1 背景

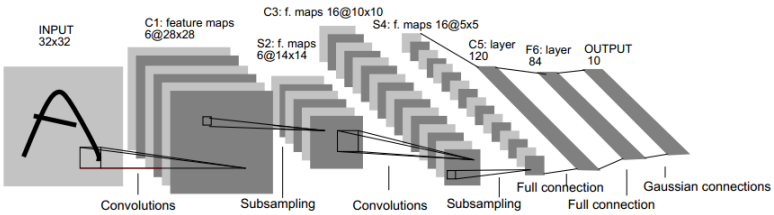


Geoffrey Hinton

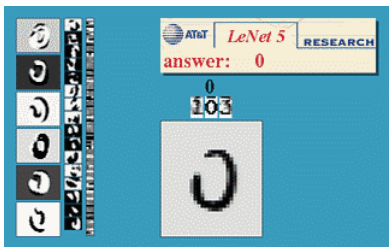


Jitendra Malik

11.1 背景



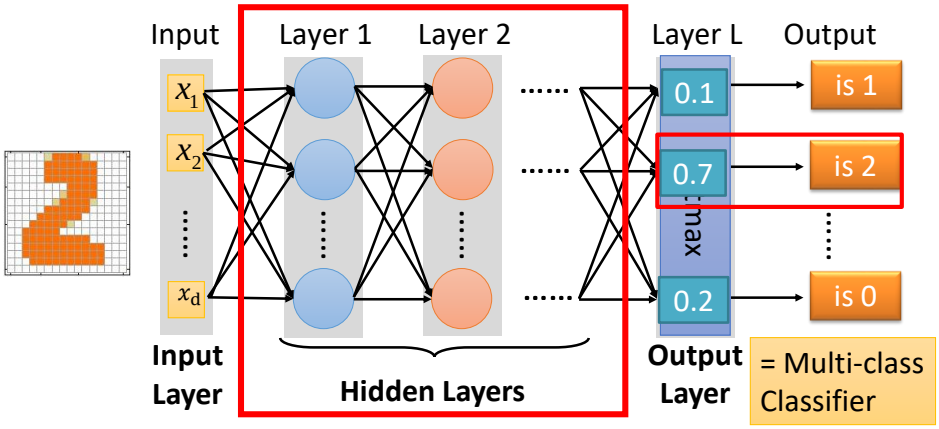
LeCun in 1989



Digit recognition, AT&T labs

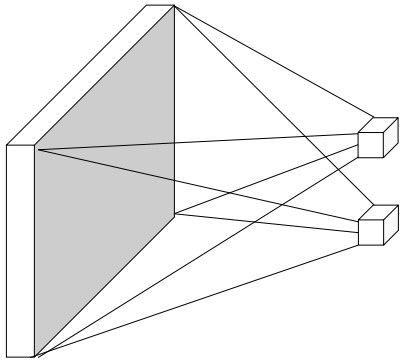
11.1 背景

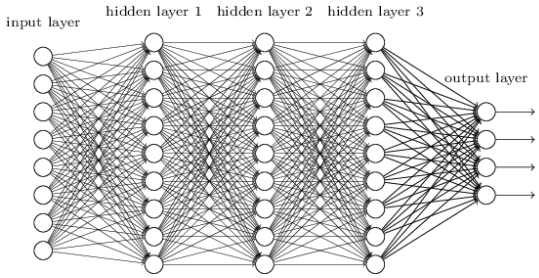
神经网络(*Neural Networks*):



11.1 背景

神经网络(*Neural Networks*):





全连接层(Fully connected layer)

Image Size in CIFAR10: 32*32*3=3072

Image Size : 200*200*3=120000

第十一讲 卷积神经网络(*Convolutional Neural Networks*)

11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

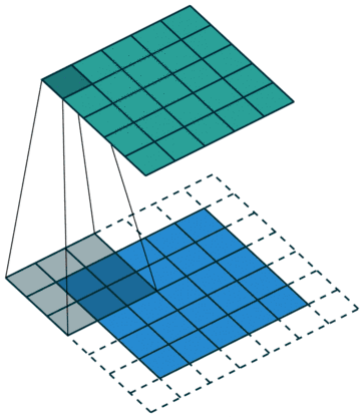
11.4 应用示例(*Case Study*)

11.2 卷积核的知识

二维卷积(*convolution in 2D*):

$$f[n,m] * h[n,m]$$

$$= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k,l]\, h[n-k, m-l]$$



11.2 卷积核的知识

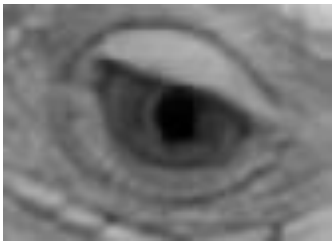
二维卷积应用示例(*convolution in 2D – examples* **):**



f[m,n]

$$* \frac{1}{9} \begin{bmatrix} \cdot 1 & \cdot 1 & \cdot 1 \\ \cdot 1 & \cdot 1 & \cdot 1 \\ \cdot 1 & \cdot 1 & \cdot 1 \end{bmatrix} =$$

h[m,n]



g[m,n]

11.2 卷积核的知识

二维卷积应用示例(*convolution in 2D – examples* **):**



f[m,n]

$\otimes$ [-1, 1] = h[m,n]



g[m,n]

11.2 卷积核的知识

二维卷积应用示例(*convolution in 2D – examples* **):**



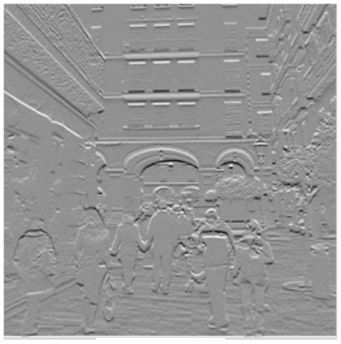
f[m,n]

$\otimes$

$[-1, 1]^T$

h[m,n]

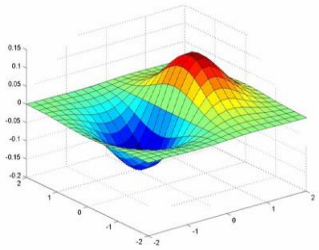
=



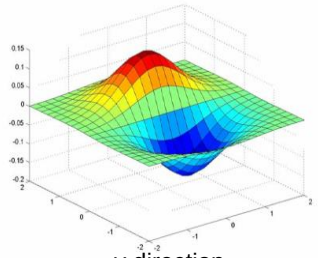
g[m,n]

11.2 卷积核的知识

二维卷积应用示例(*convolution in 2D – examples* **):**

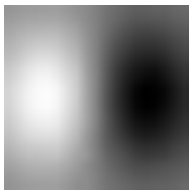


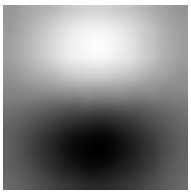
x-direction



y-direction

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

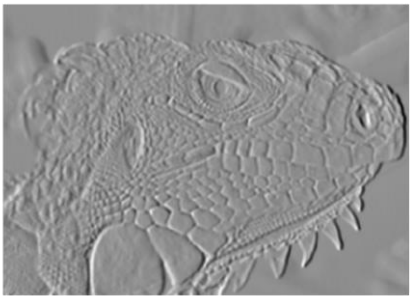




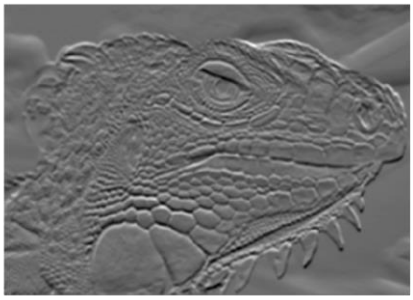
$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

11.2 卷积核的知识

二维卷积应用示例(*convolution in 2D – examples* **):**



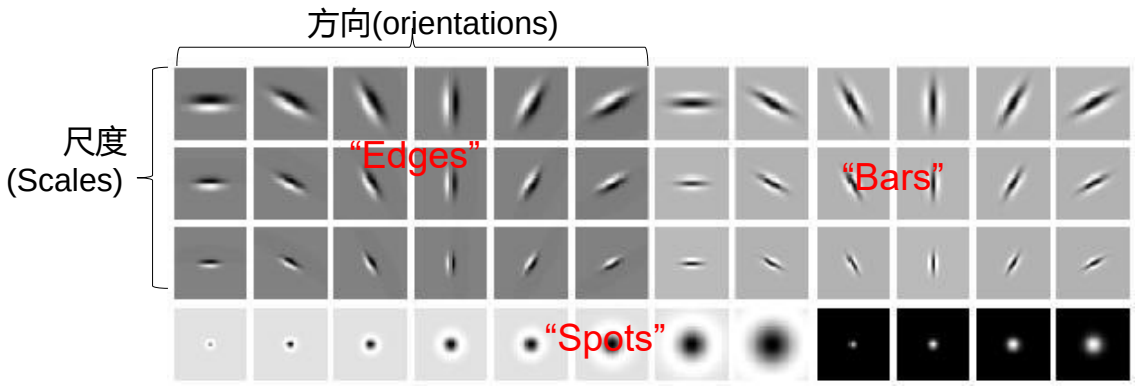
X-Derivative of Gaussian



Y-Derivative of Gaussian

11.2 卷积核的知识

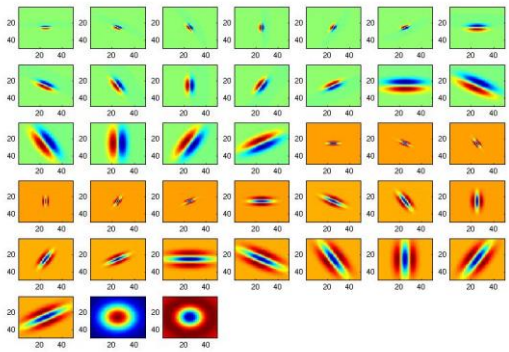
滤波器组(*Filter banks***):**



11.2 卷积核的知识

滤波器组(*Filter banks*):

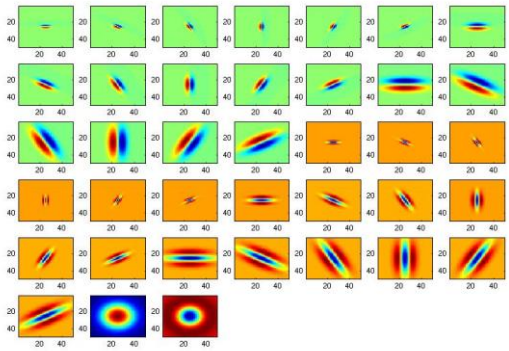


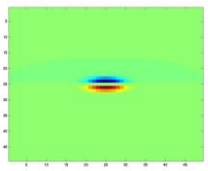


11.2 卷积核的知识

滤波器组(*Filter banks*):



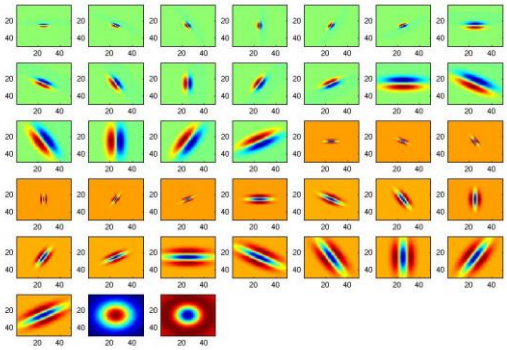


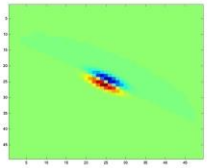


11.2 卷积核的知识

滤波器组(*Filter banks*):





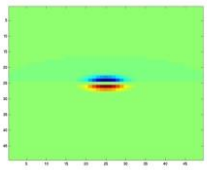


11.2 卷积核的知识

滤波器组(*Filter banks*):



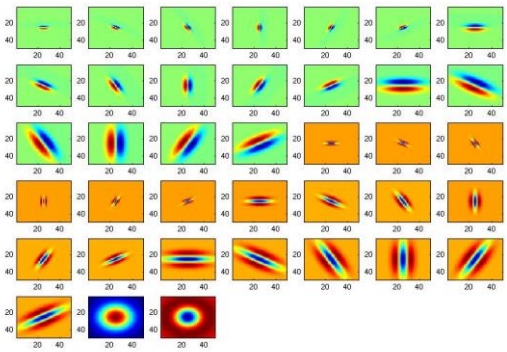


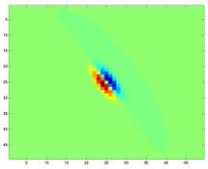


11.2 卷积核的知识

滤波器组(*Filter banks*):

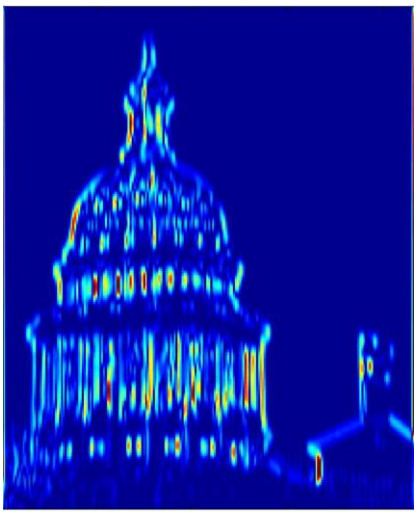


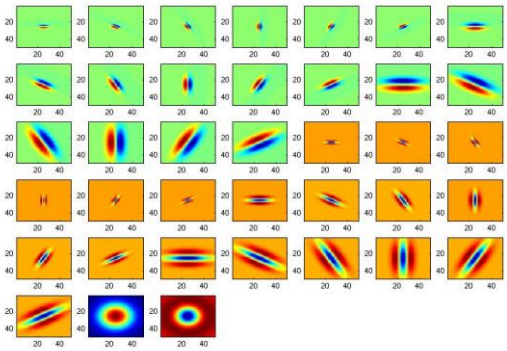


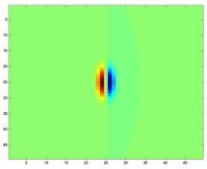


11.2 卷积核的知识

滤波器组(*Filter banks*):



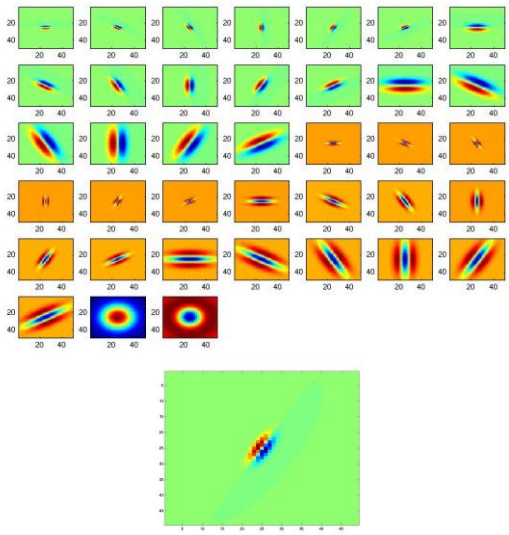




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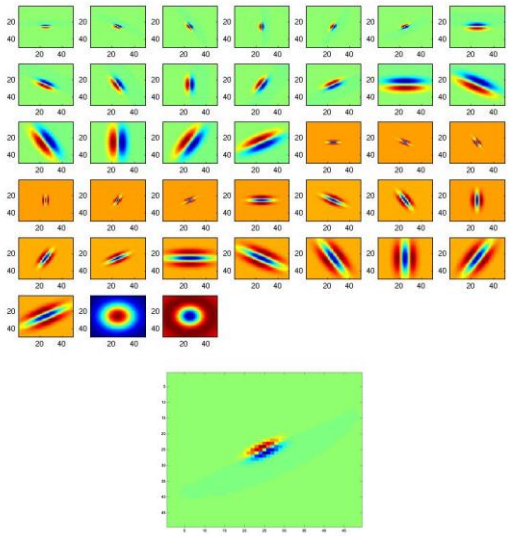




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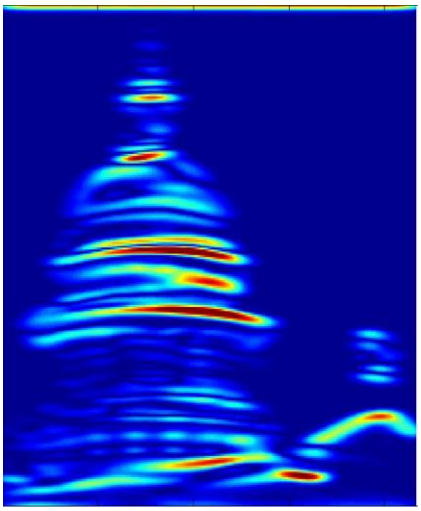
滤波器组(*Filter banks*):

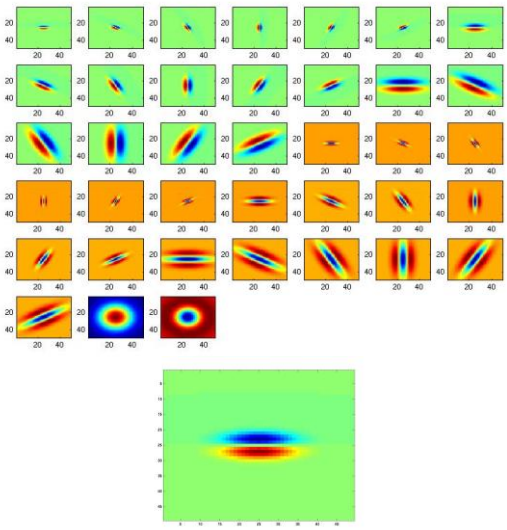




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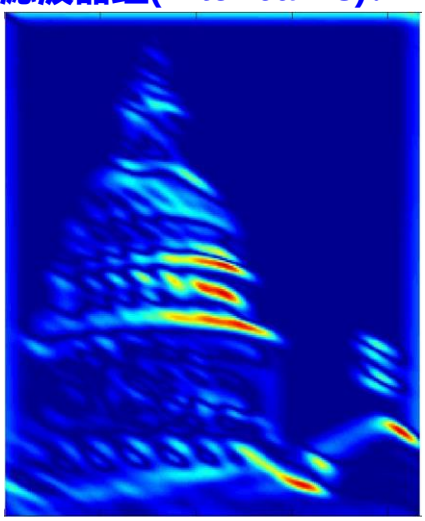
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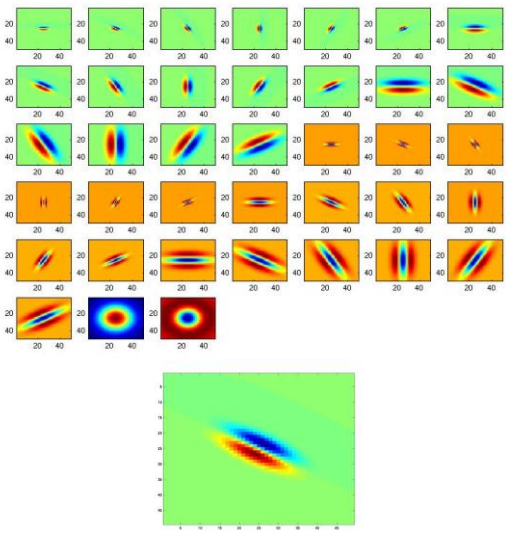




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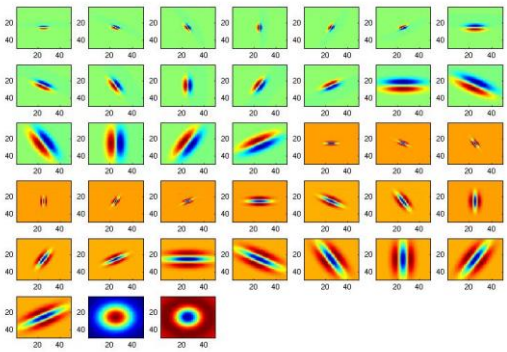


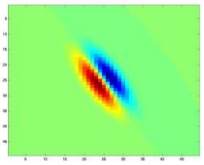


11.2 卷积核的知识

滤波器组(*Filter banks*):

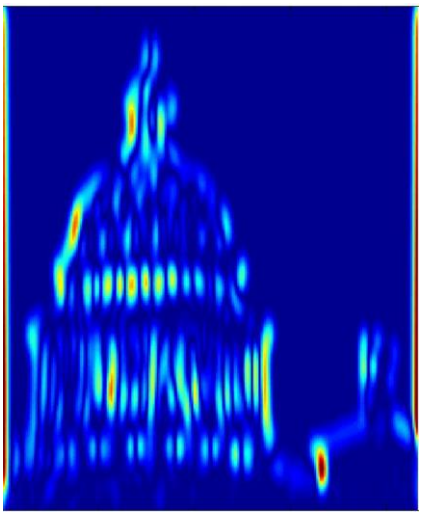


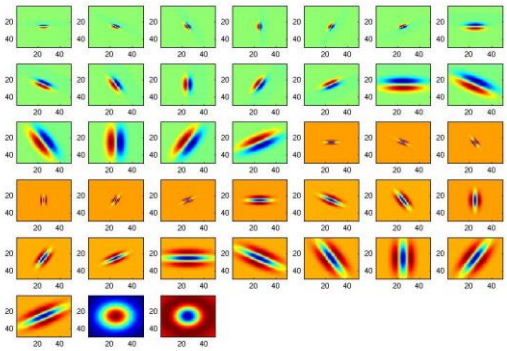


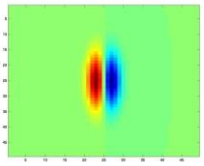


11.2 卷积核的知识

滤波器组(*Filter banks*):



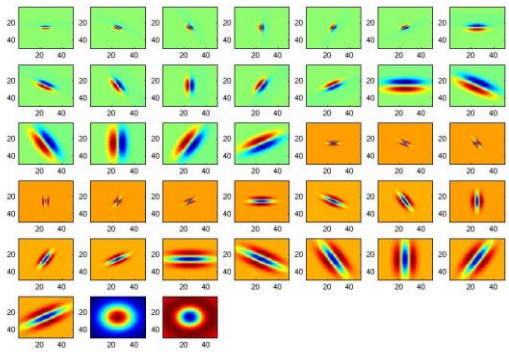


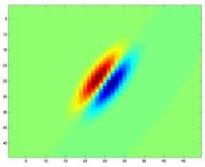


11.2 卷积核的知识

滤波器组(*Filter banks*):

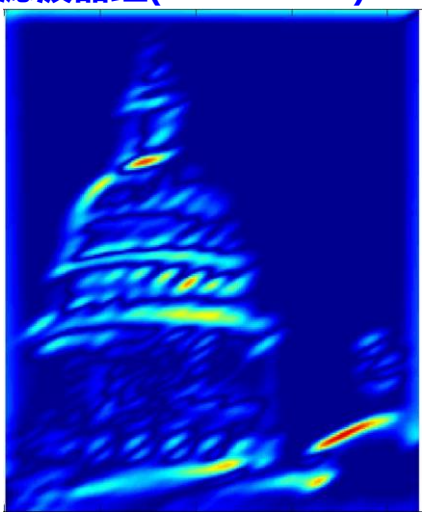


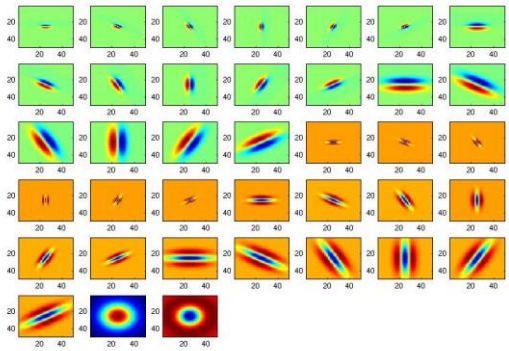


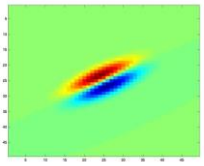


11.2 卷积核的知识

滤波器组(*Filter banks*):

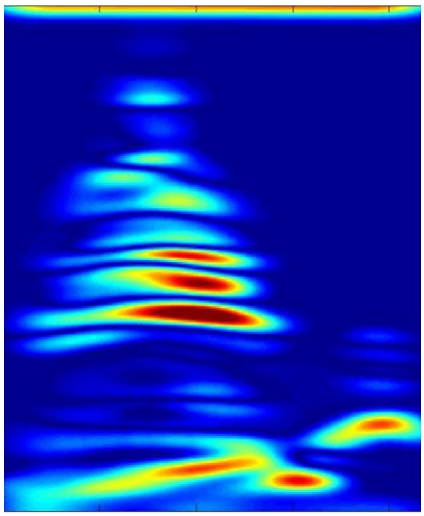


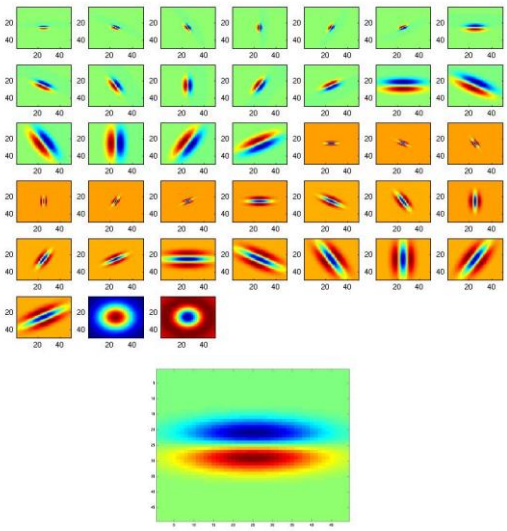




11.2 卷积核的知识

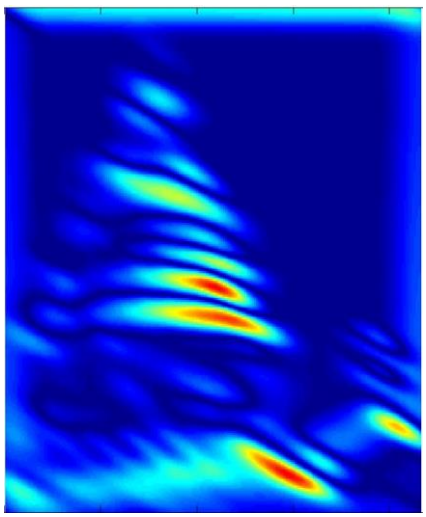
滤波器组(*Filter banks*):

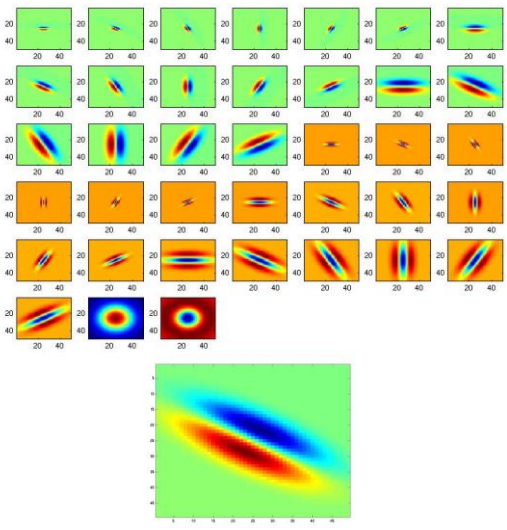




11.2 卷积核的知识

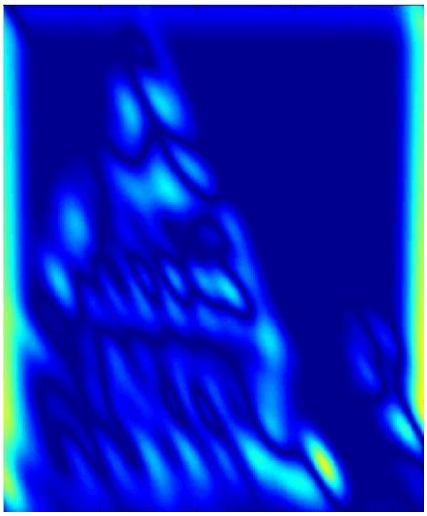
滤波器组(*Filter banks*):

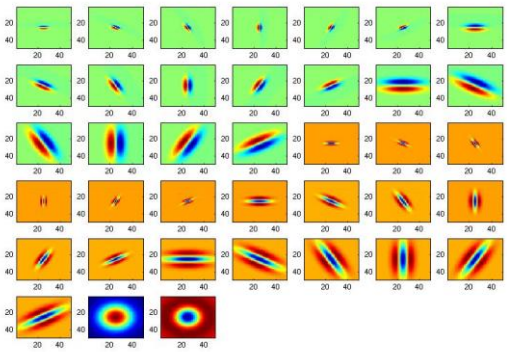


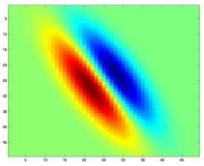


11.2 卷积核的知识

滤波器组(*Filter banks*):

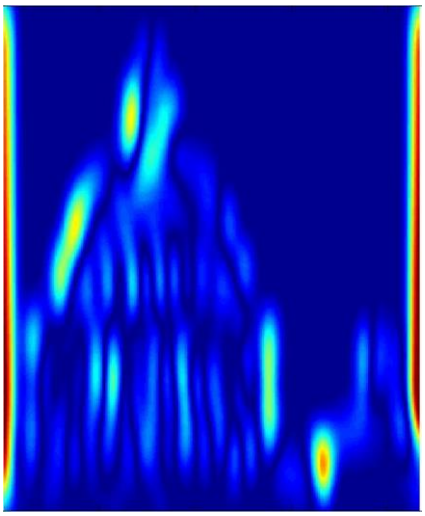


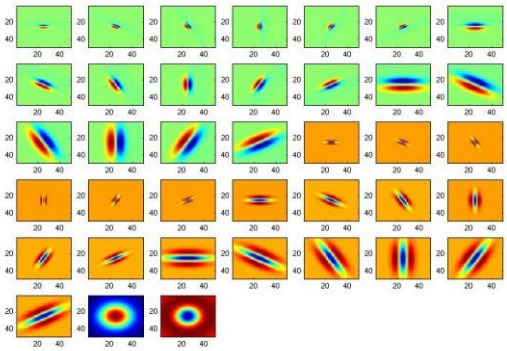


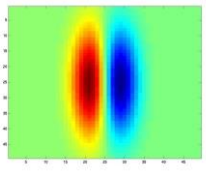


11.2 卷积核的知识

滤波器组(*Filter banks*):

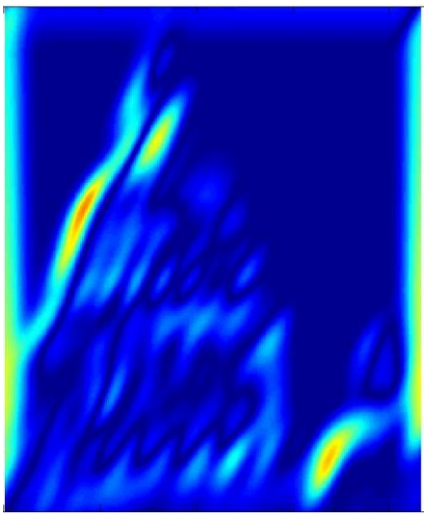


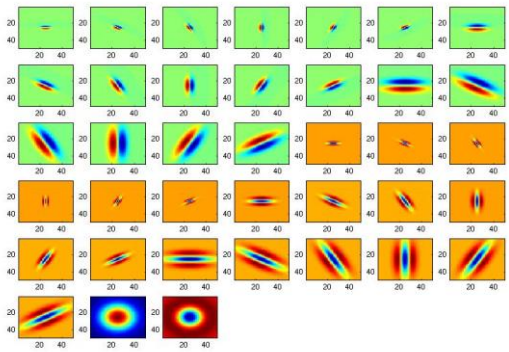


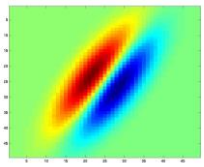


11.2 卷积核的知识

滤波器组(*Filter banks*):

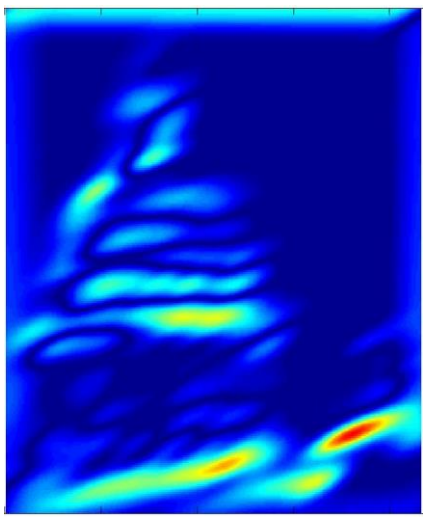


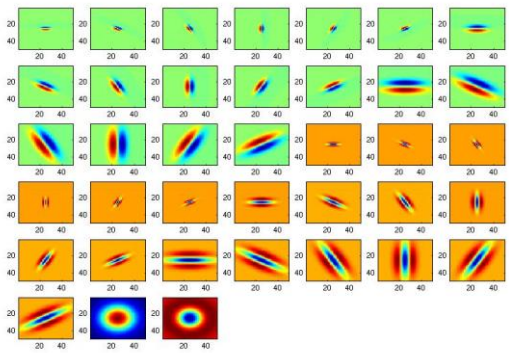


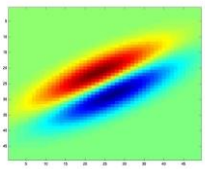


11.2 卷积核的知识

滤波器组(*Filter banks*):



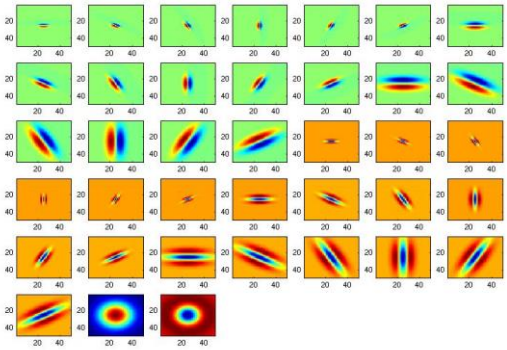


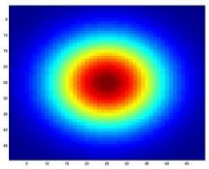


11.2 卷积核的知识

滤波器组(*Filter banks*):



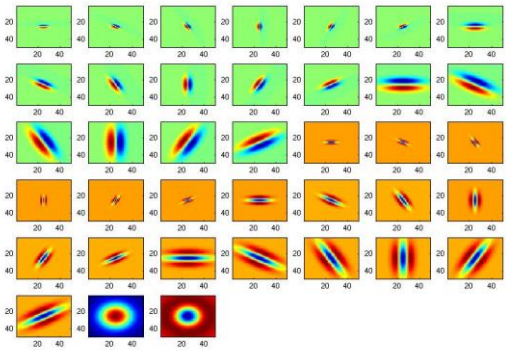




11.2 卷积核的知识

滤波器组(*Filter banks*):





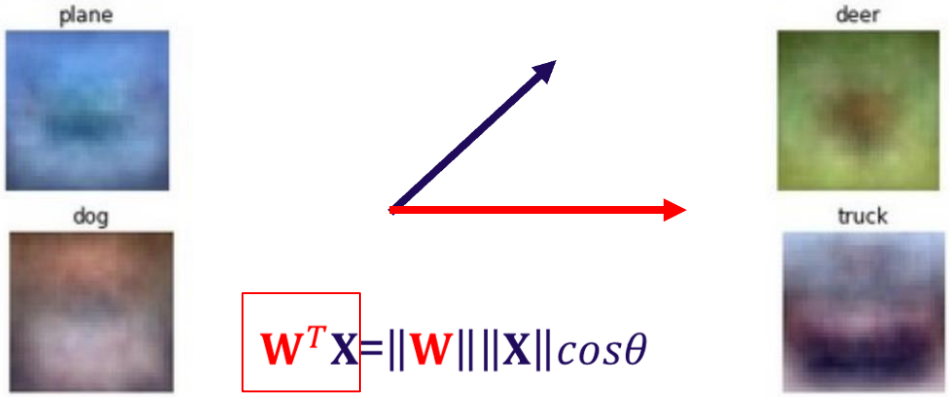
[r1, r2, …, r38]

每个像素上由滤波器组产生一系列响应构造成特征向量

2.1 感知器模型参数空间

感知器模型的可视化：

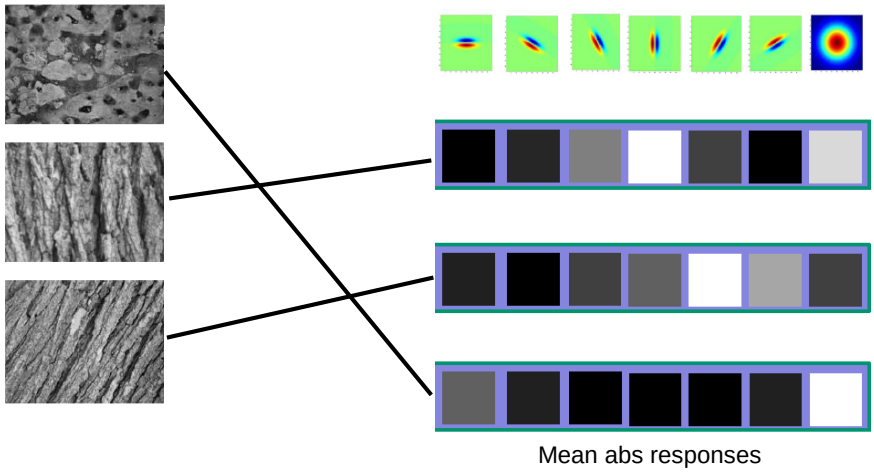
$$f[n,m] * h[n,m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k,l]\, h[n-k, m-l]$$



$$\mathbf{W}^T\mathbf{X} = \|\mathbf{W}\|\|\mathbf{X}\| cos\theta$$

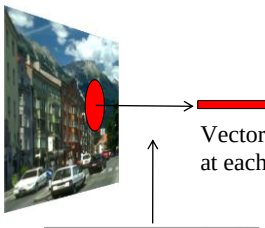
11.2 卷积核的知识

滤波器组(*Filter banks*)：



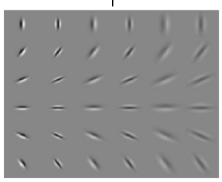
11.2 卷积核的知识

滤波器组(*Filter banks*):



Vector of filter responses at each pixel

Kmeans over a set of vectors on a collection of images

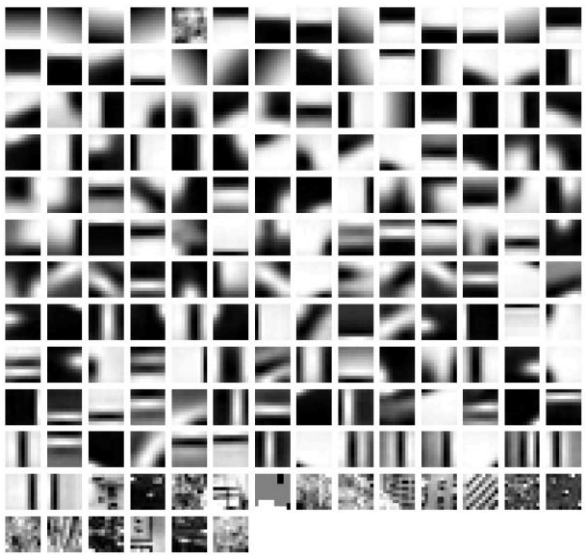


Filter bank

Textons

11.2 卷积核的知识

Codewords dictionary formation

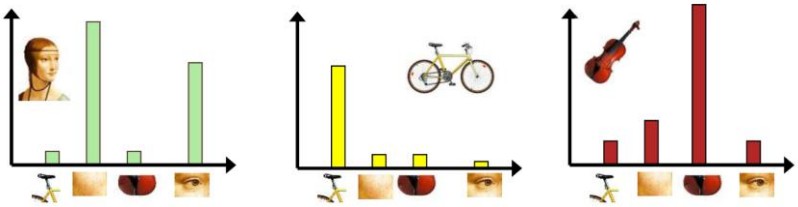


11.2 卷积核的知识

1. Extract features
2. Learn “visual vocabulary”
3. Represent images by frequencies of “visual words”



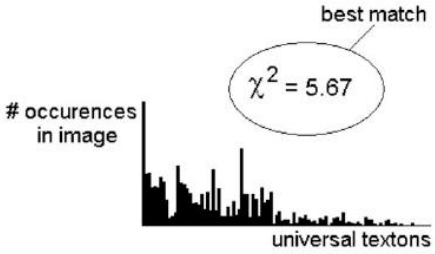
Bag of words



11.2 卷积核的知识

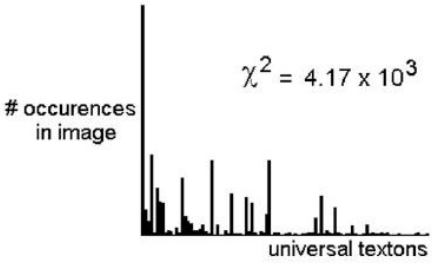


label = bedroom





label = beach



第十一讲 卷积神经网络(*Convolutional Neural Networks*)

11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

11.4 应用示例(*Case Study*)

11.3 卷积神经网络

图像/视觉中采用CNN结构的原因

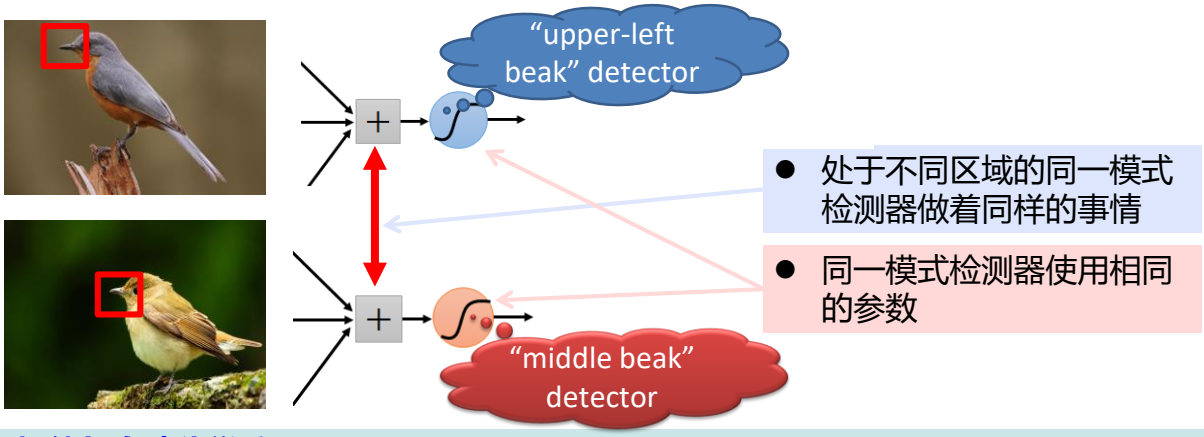
- 反映物体特性的模式往往只占图像中很小的一部分
  - 为了发现这些模式，神经元无需与图像中的所有像素连接
  - 神经元与图像中局部区域像素连接，所需的参数量会更少



11.3 卷积神经网络

图像/视觉中采用CNN结构的原因

- 物体同样的模式可能出现在图像中的不同区域

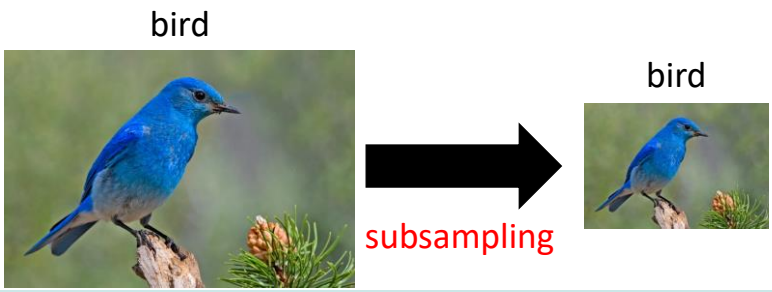


- 处于不同区域的同一模式检测器做着同样的事情
- 同一模式检测器使用相同的参数

11.3 卷积神经网络

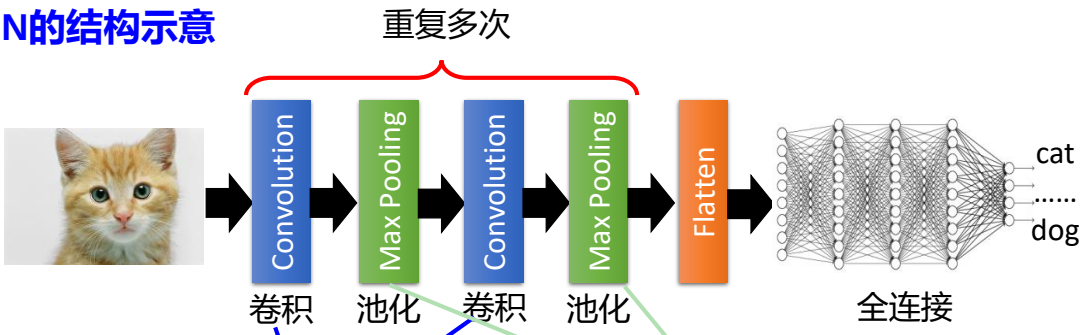
图像/视觉中采用CNN结构的原因

- 下采样不会改变物体的类别属性
  - 下采样降低了图像大小
  - 网络处理较小的图像需要的参数量也随之减少



11.3 卷积神经网络

CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

特性2

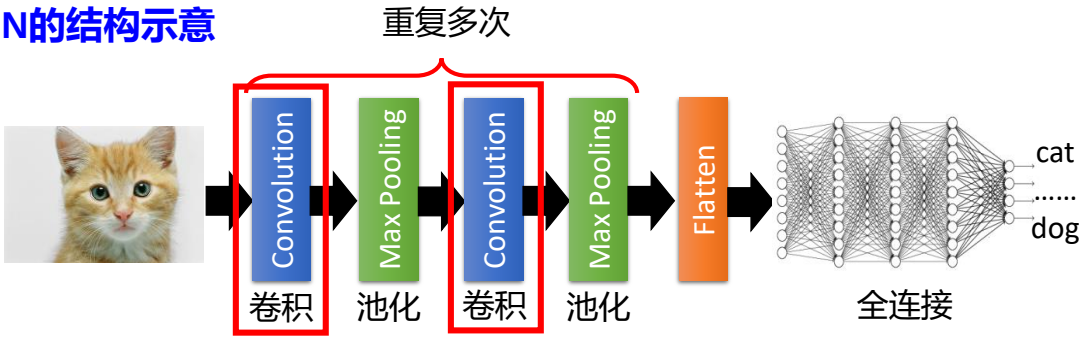
- 物体同样的模式可能出现在图像中的不同区域

特性3

- 下采样不会改变物体的类别属性

11.3 卷积神经网络

CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

特性2

- 物体同样的模式可能出现在图像中的不同区域

特性3

- 下采样不会改变物体的类别属性

11.3 卷积神经网络

CNN—卷积层工作过程介绍

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

Matrix

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Matrix

⋮

特性1

- 每个滤波器用于检测一种模式

滤波器是可通过学习获得的网络参数

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3 -1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

特性2

- 相同的模式处于图像的不同位置，可以用参数相同的检测器

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

-1	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

-1	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

-1	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

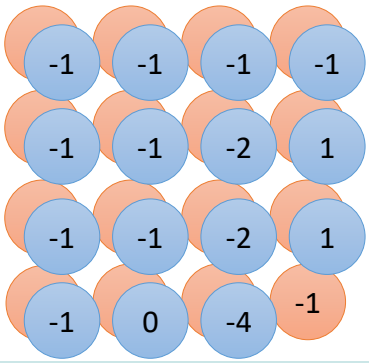
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

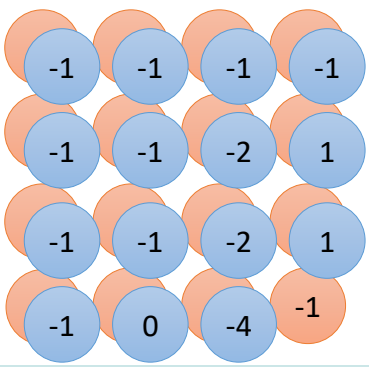
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

11.3 卷积神经网络

CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

所有滤波器做同样的操作

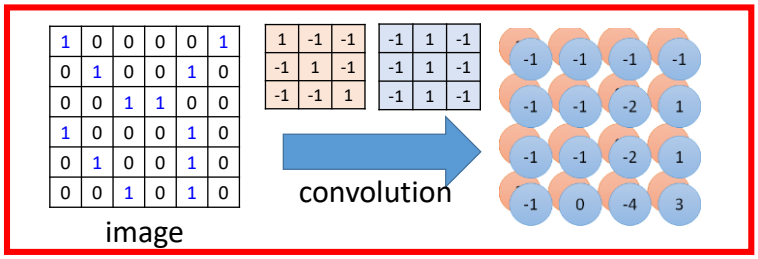
-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

-1	-1	-1	-1
	特征图 Feature Map		1
-1	-1	-2	1
-1	0	-4	3

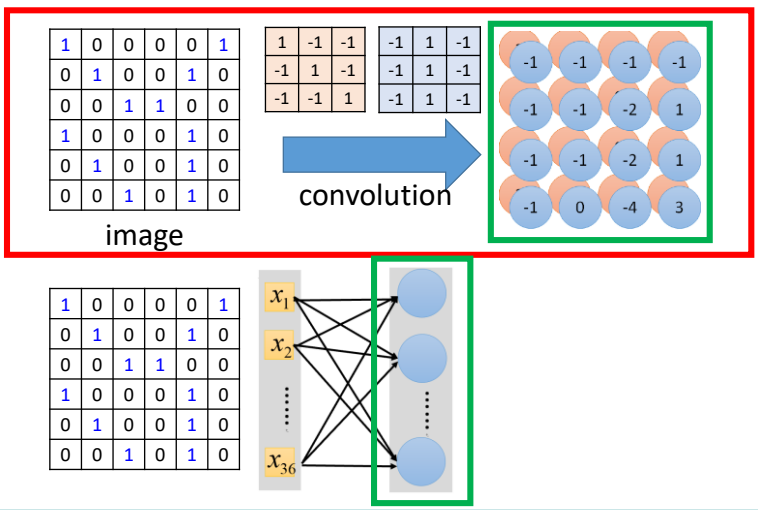
11.3 卷积神经网络

CNN—卷积层



11.3 卷积神经网络

CNN—卷积层与全连接层



11.3 卷积神经网络

1	-1	-1
-1	1	-1
-1	-1	1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1: 1
2: 0
3: 0
4: 0
⋮
7: 0
8: 1
9: 0
10: 0
⋮
13: 0
14: 0
15: 1
16: 1

11.3 卷积神经网络

1	-1	-1
-1	1	-1
-1	-1	1

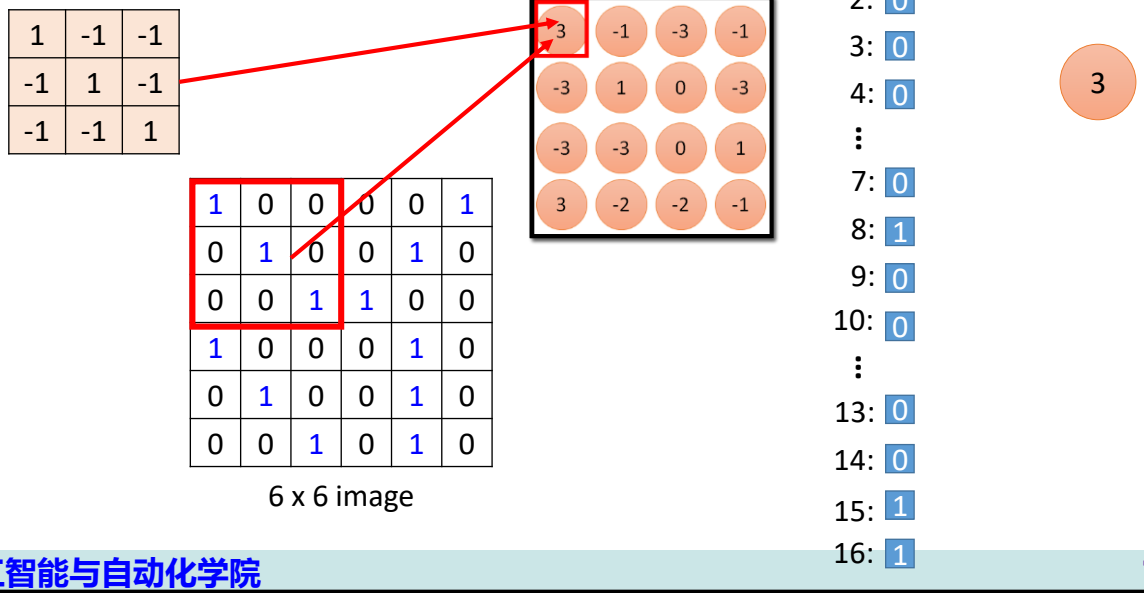
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

1: 1
2: 0
3: 0
4: 0
⋮
7: 0
8: 1
9: 0
10: 0
⋮
13: 0
14: 0
15: 1
16: 1

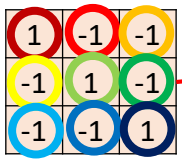
11.3 卷积神经网络

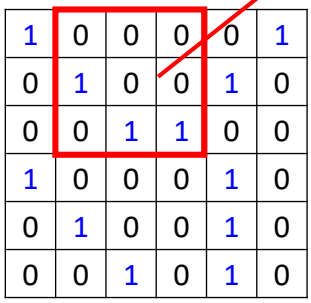


11.3 卷积神经网络

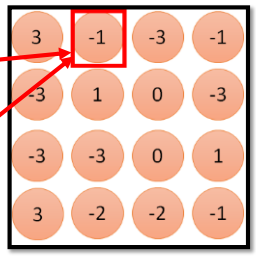


11.3 卷积神经网络





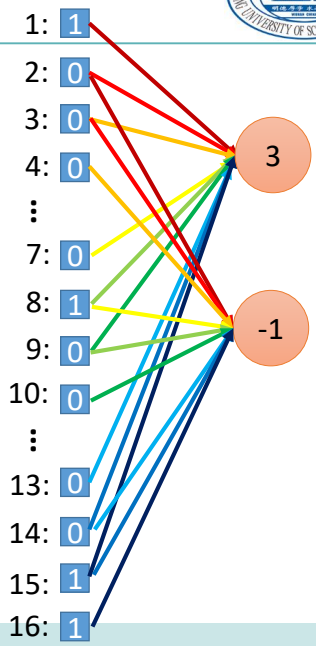
6 x 6 image



仅仅需要9个参数，而不是全连接

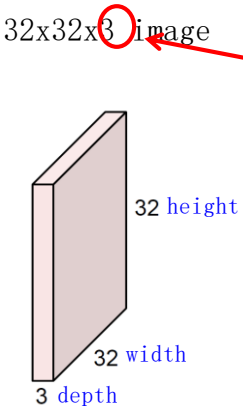
参数量减少

甚至可以更少

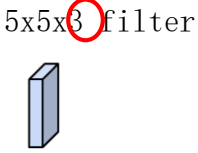


11.3 卷积神经网络

彩色图像的卷积层

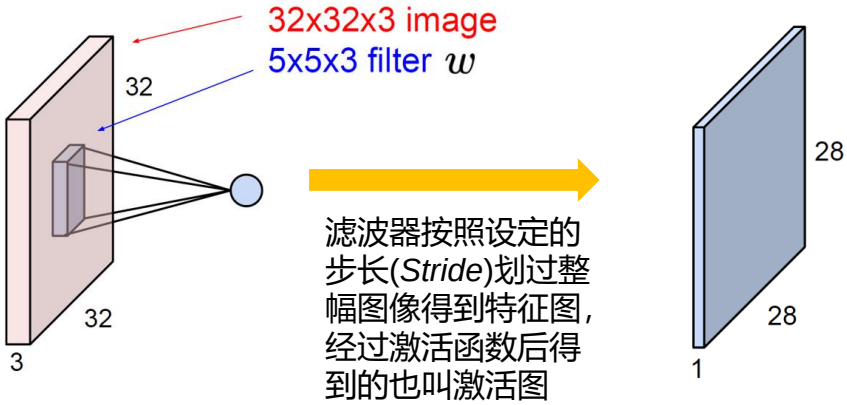


滤波器的通道数要与输入数据的通道数保持一致



11.3 卷积神经网络

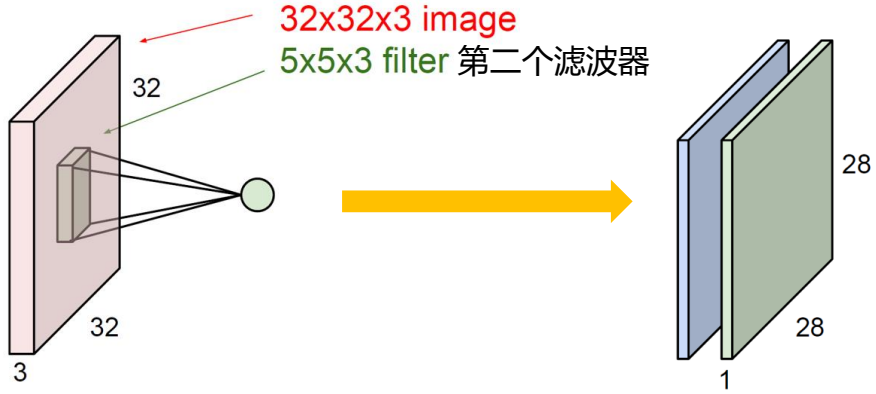
彩色图像的卷积层



滤波器按照设定的步长(*Stride*)划过整幅图像得到特征图，经过激活函数后得到的也叫激活图(*activation map*)

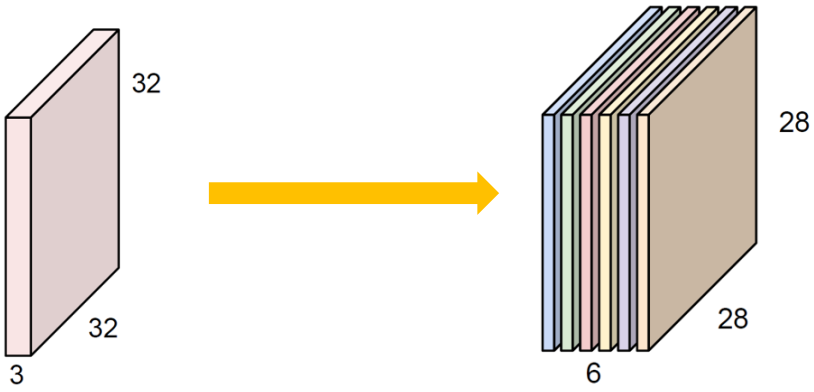
11.3 卷积神经网络

彩色图像的卷积层



11.3 卷积神经网络

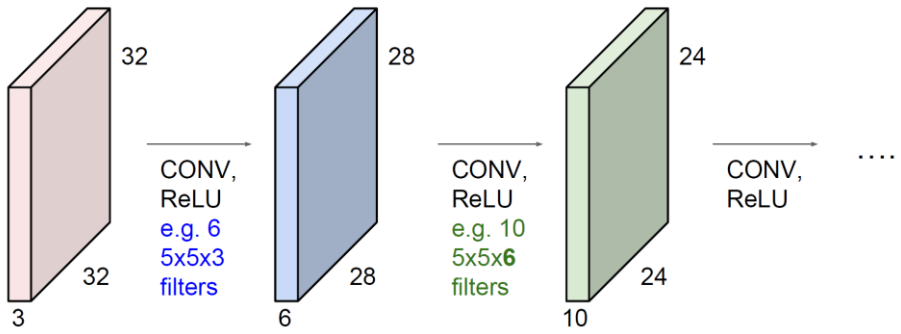
彩色图像的卷积层



- 如果有6个5*5*3的滤波器，就可以分别得到6个特征图(激活图)
- 6个特征图堆叠在一起就构成了一幅28*28*6的“新图像”

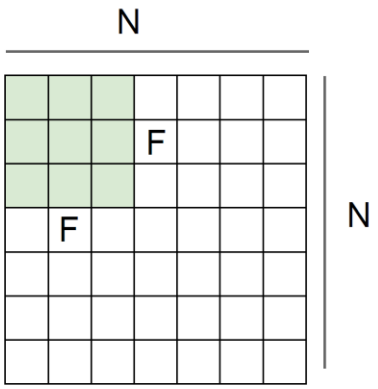
11.3 卷积神经网络

彩色图像的一系列卷积层操作



11.3 卷积神经网络

步长(*Stride*)、卷积后的图像大小

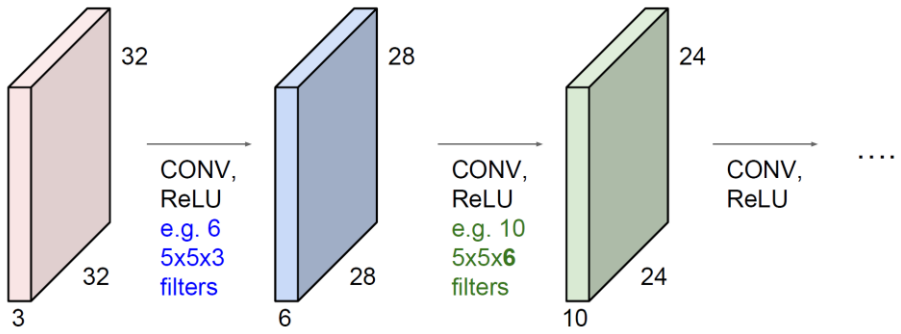


Output size:
(N - F) / stride + 1

e.g. N = 7, F = 3:
stride 1 => (7 - 3)/1 + 1 = 5
stride 2 => (7 - 3)/2 + 1 = 3
~~stride 3 => (7 - 3)/3 + 1 = 2.33~~

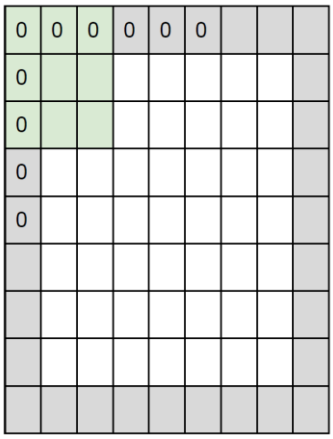
11.3 卷积神经网络

步长(*Stride*)、卷积后的图像大小



11.3 卷积神经网络

填充(*Padding*)以保持卷积后的图像大小不变



- 实际应用中，填充常常就是在边界处补0

  Input: 7*7; Filter: 3*3; Stride: 1; Pad: 1

  Output: 7*7; ((*N-F*)/*S+1*)

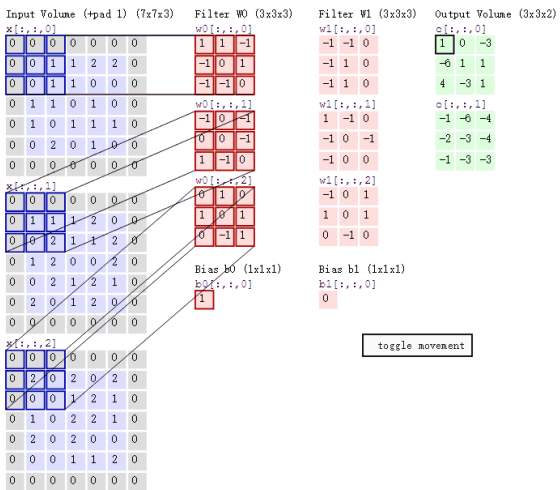
- 保证输出与输入尺寸一样时，确定填充值的一般表达式：

  Input: *N*N*; Filter: *F*F*; Stride: *S*;

  Pad: *P=(N(S-1)+F-S)/2*

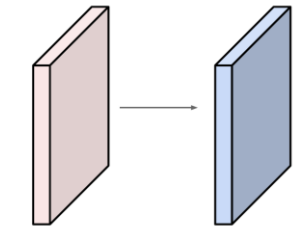
11.3 卷积神经网络

彩色图像、步长为2的卷积过程：



11.3 卷积神经网络

卷积层的模型参数量计算示例：



输入层：　图像大小为32*32*3

卷积计算：　10个5*5*3的滤波器，Stride为1，Pad为2

卷积层：　卷积后得到的特征图大小是多少

1个滤波器：(32+2*2-5)/1+1 = 32

10个滤波器：32*32*10

连接输入与卷积层的参数量是多少

1个滤波器：5*5*3+1 = 76

10个滤波器：10*76 = 760

11.3 卷积神经网络

卷积层的模型参数量计算总结：

输入大小为$W_1 \times H_1 \times D_1$，　滤波器个数为$K$，尺寸为$F$，*Stride*为$S$，零填充为$P$

卷积后得到的特征图大小是$W_2 \times H_2 \times D_2$，　这里：

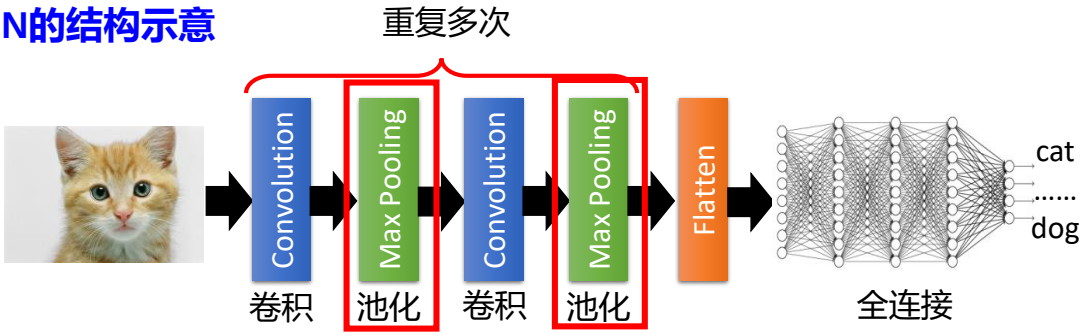
$$W_2 = \frac{W_1 - F + 2P}{S} + 1, \quad H_2 = \frac{H_1 - F + 2P}{S} + 1, \quad D_2 = K$$

这一层的参数量是：

$$(F \cdot F \cdot D_1) \cdot K + K$$

11.3 卷积神经网络

CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

特性2

- 物体同样的模式可能出现在图像中的不同区域

特性3

- 下采样不会改变物体的类别属性

11.3 卷积神经网络

CNN – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

-1	1	-1
-1	1	-1
-1	1	-1

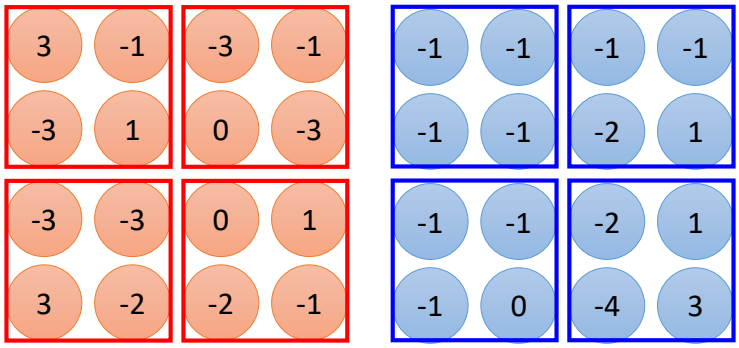
Filter 2

11.3 卷积神经网络

CNN – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

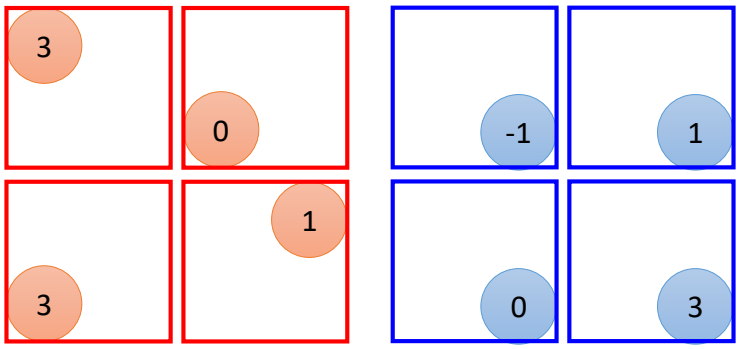
按照步长为2，2×2 的尺寸做池化(*Pooing*)

11.3 卷积神经网络

CNN – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

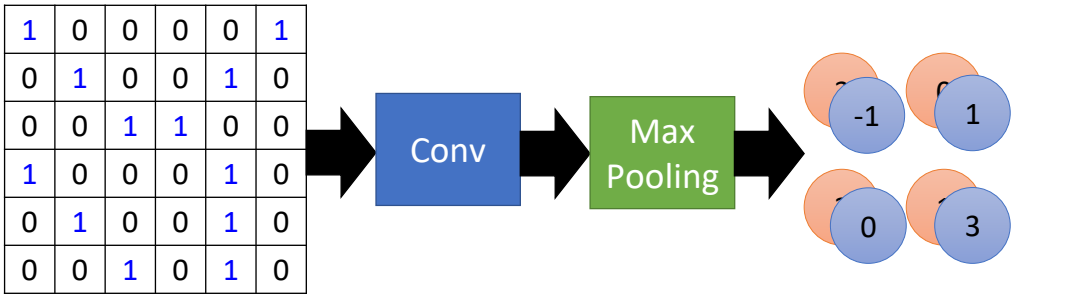


-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

11.3 卷积神经网络

CNN – Max Pooling



$2 \times 2 \times 2$ “新图像”

11.3 卷积神经网络

池化层的模型参数量计算总结：

输入大小为$W_1 \times H_1 \times D_1$， 池化(*Pooling*)窗口的尺寸为$F$，步长(*Stride*)为$S$

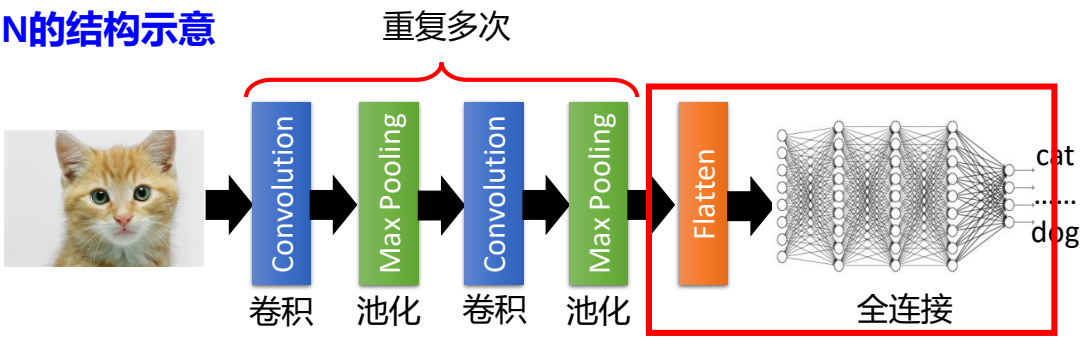
池化后得到的特征图大小是$W_2 \times H_2 \times D_2$， 这里：

$$W_2 = \frac{W_1 - F}{S} + 1, \qquad H_2 = \frac{H_1 - F}{S} + 1, \qquad D_2 = D_1$$

池化层的参数量是：0

11.3 卷积神经网络

CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

特性2

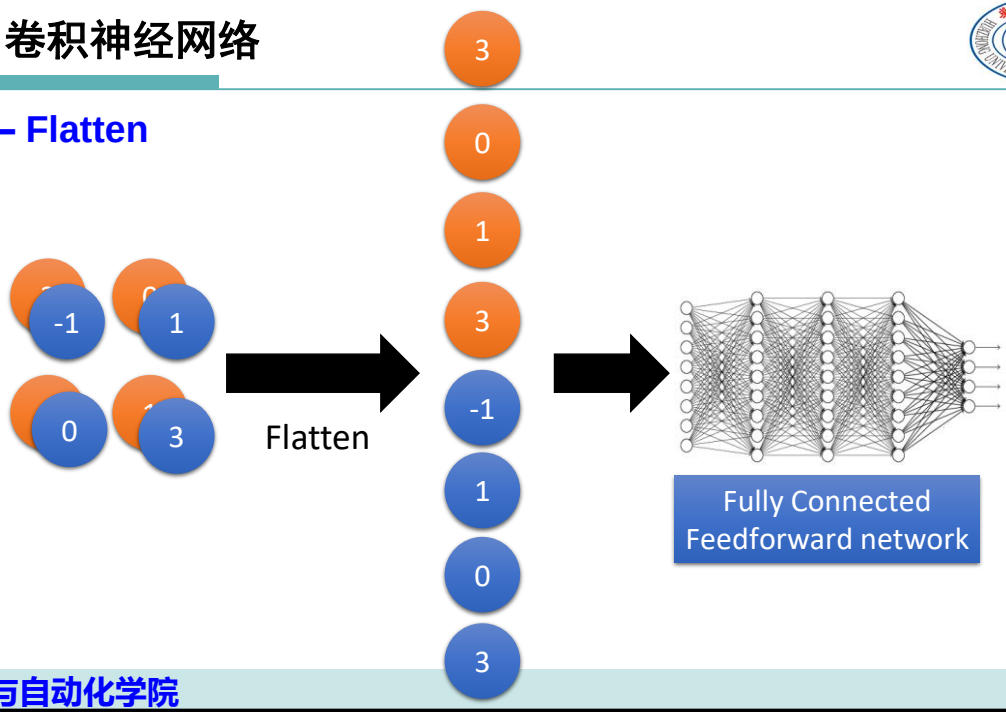
- 物体同样的模式可能出现在图像中的不同区域

特性3

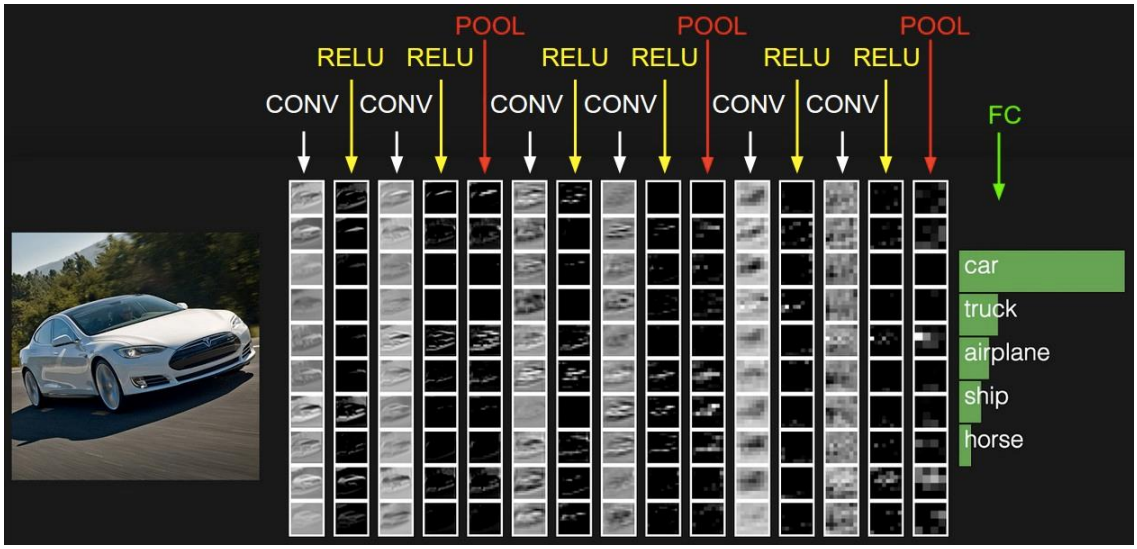
- 下采样不会改变物体的类别属性

11.3 卷积神经网络

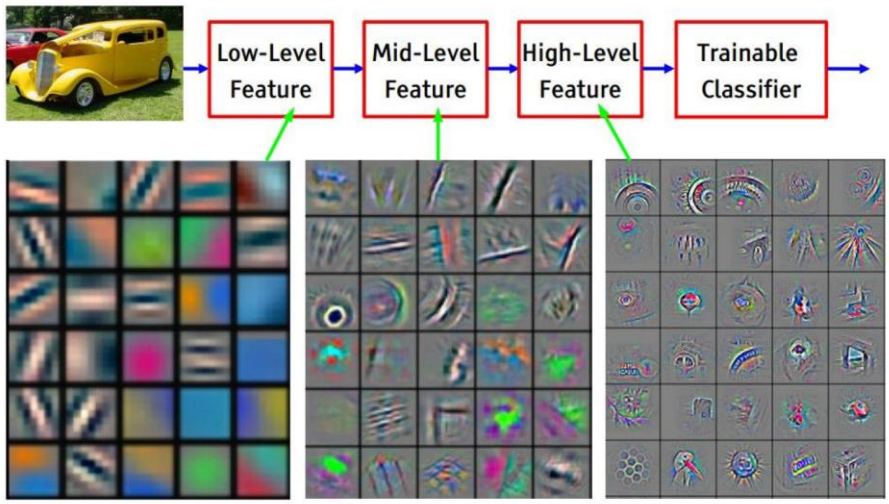
CNN – Flatten



11.3 卷积神经网络



11.3 卷积神经网络



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

所有参数都是通过学习得到

端到端的学习

第十一讲 卷积神经网络(*Convolutional Neural Networks*)

11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

11.4 应用示例(*Case Study*)

CNN结构演化-早期尝试

- **1998-LeNet**

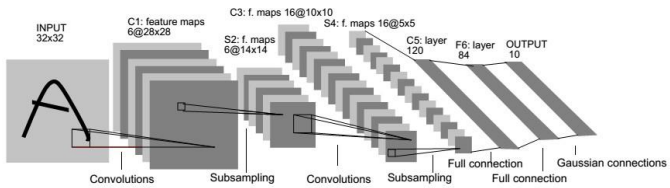


Fig. 2. Architecture of LeNet-5, a Convolutional Neural Network, here for digits recognition. Each plane is a feature map, i.e. a set of units whose weights are constrained to be identical.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	X				X	X	X			X	X	X	X		X	X
1	X	X				X	X	X			X	X	X	X		X
2	X	X	X				X	X	X			X		X	X	X
3		X	X	X			X	X	X	X			X		X	X
4			X	X	X			X	X	X	X			X	X	
5				X	X	X			X	X	X	X			X	X

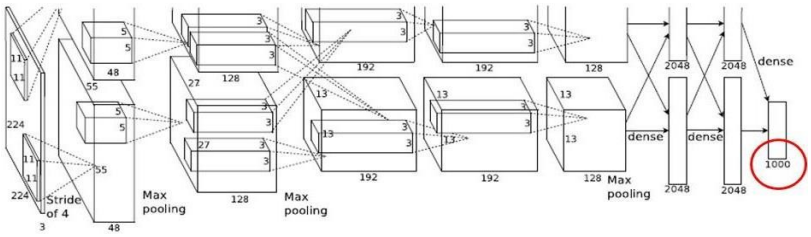
TABLE I

EACH COLUMN INDICATES WHICH FEATURE MAP IN S2 ARE COMBINED BY THE UNITS IN A PARTICULAR FEATURE MAP OF C3.

CNN早期应用，手写字符识别准确率达95%以上

CNN结构演化-历史突破

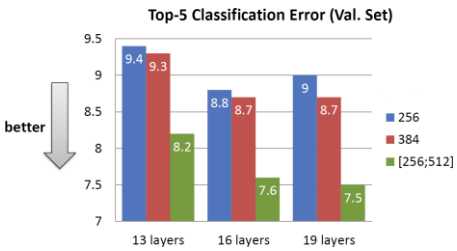
- **2012-AlexNet**
  - 非线性激活函数：ReLU
  - 防止过拟合：Dropout, 数据增广
  - 大数据训练：百万级ImageNet图像数据
  - 其他：分Group实现双GPU并行，LRN归一化层



ImageNet分类准确率达到，1000多类，84.7%（Top5）

CNN结构演化-网络加深

- **2014-VGG Net**
  - Slower to be deeper
    - 卷积核大小3x3
    - 卷积采样间隔1x1
    - Max Pooling间隔2x2
  - 多尺度融合

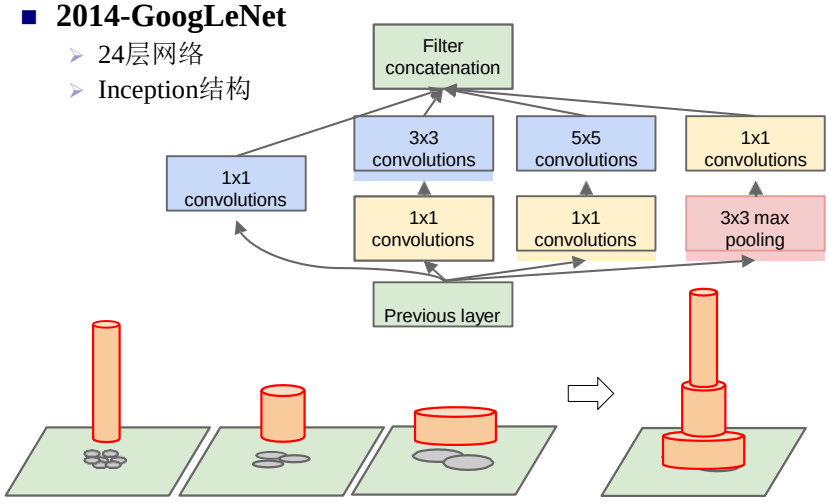




ImageNet分类准确率达到，1000多类，92.7%（Top5），16层

CNN结构演化—网络加深

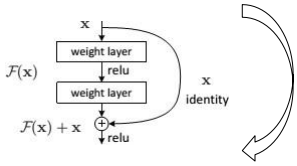
- **2014-GoogLeNet**
  - 24层网络
  - Inception结构



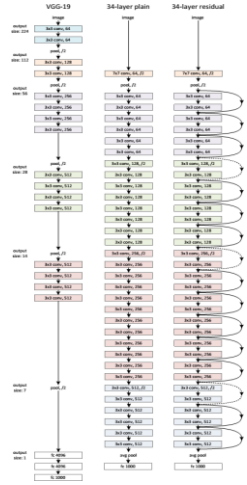
ImageNet分类准确率，1000多类，93.3%（Top5），24层

CNN结构演化—集成路线

- **2015-Deep Residual Net**



layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9



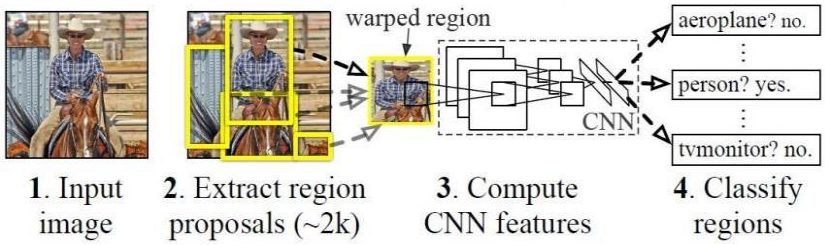
ImageNet分类准确率达到，1000多类，96.4%（Top5），152层

CNN结构演化—从分类到检测

R-CNN Ross Girshick' CVPR14

Fast R-CNN Ross Girshick ICCV15

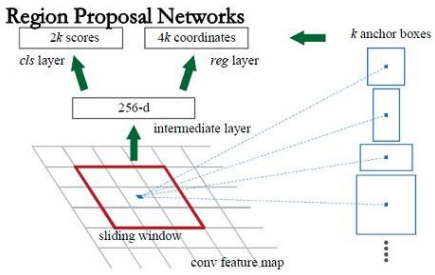
- 利用Spatial Pyramid Pooling （SPP）进行加速，整张图像只需要通过卷积网络一次
- 分类+窗口回归在统一框架中训练，不需要保留中间结果
- 速度从RCNN的46s加速到2~3s（主要时间花在region proposal)



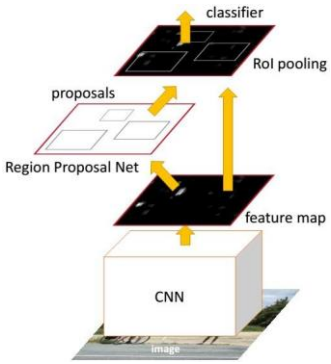
CNN结构演化—分类到检测

Faster R-CNN Shaoqing Ren' NIPS15

- **Faster R-CNN = RPN + Fast R-CNN（二阶段网络）**
- 用RPN网络代替Edge box等Region Proposal 方法，共享了RPN和Fast R-CNN卷积层参数，支持端到端训练
- 速度显著提高，GPU 200ms /张图像



候选区提取阶段（FPN）

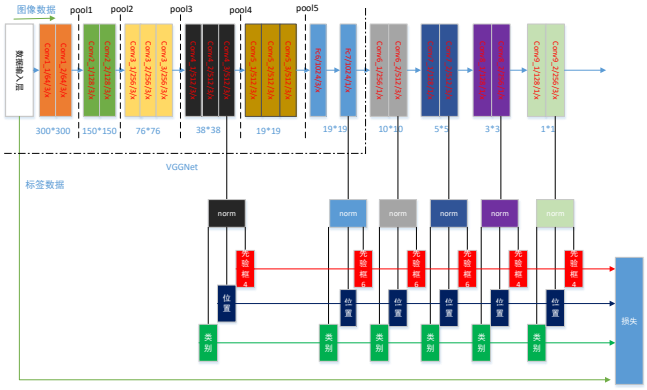


位置精调与分类阶段

CNN结构演化—分类到检测

SSD Liu'ECCV16

- 一阶段网络，融合候选区域选取和分类
- 运用多个特征图进行预测，提高多尺度适应能力
- 速度进一步提高，GPU可做到实时处理



CNN结构演化—分割分类

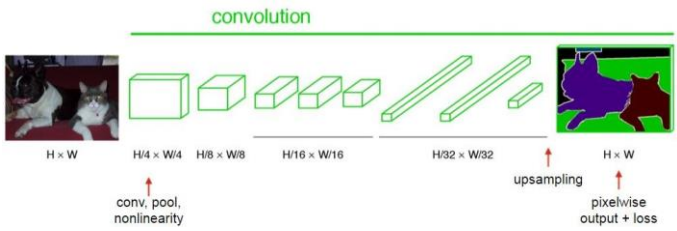
FCN Jon Long's CVPR15

- 结构特点：

1）支持任意长宽比图像，使用1x1卷积核替代全连接保持空间关系

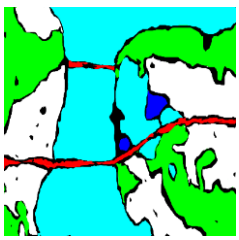
2）利用反卷积层进行上采样

3）对输入图像的每个像素进行分类（Softmax Loss)



- 训练机制：基于ImageNet图像分类模型fine-tune, base_lr一般需较小，例如10^{-6}



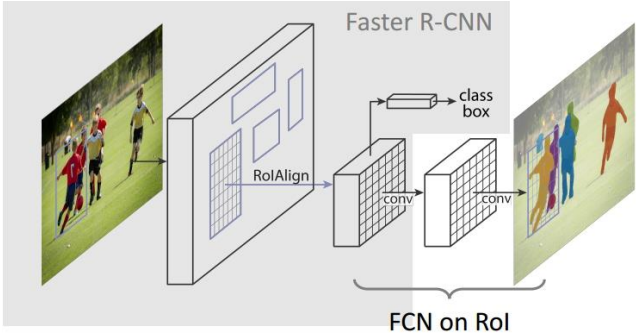


FCN遥感影像分割

CNN结构演化—分割与检测的结合,实例分割

Mask R-CNN Shaoqing Ren' NIPS15

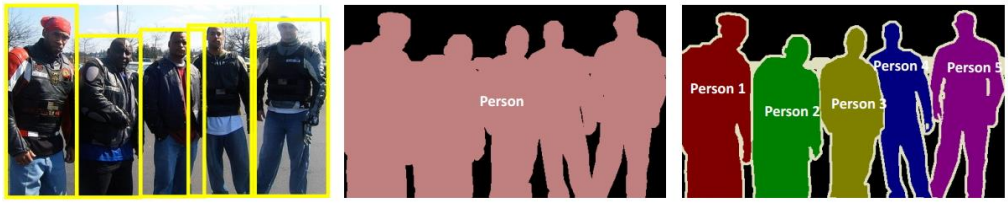
- **Mask R-CNN = Faster R-CNN + FCN**
- 将检测与分割任务进行融合，提出了实例分割的思想
- 目标检测准确率再次提高
- 可用来进行目标检测，分割，和人物关键点检测等



CNN结构演化—分割与检测的结合,实例分割

Mask R-CNN Shaoqing Ren' NIPS15

- 目标检测，语义分割，实例分割的区别与联系



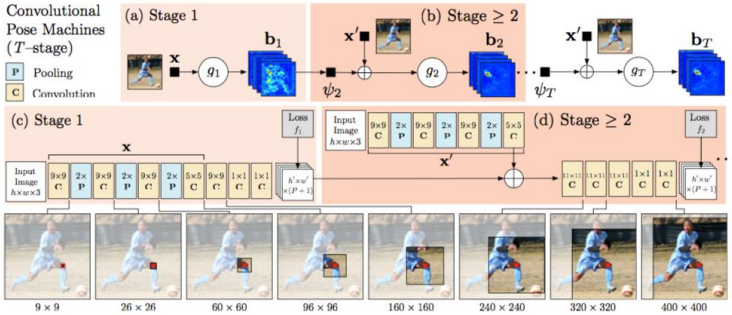
Object Detection　　Semantic Segmentation　　**Instance Segmentation**

目标检测+语义分割→实例分割

CNN结构演化—检测任务的再次升级，姿态检测

OpenPose SE WEI'CVPR16

- 用各部件响应图来表达各部件之间的空间约束
- 网络分为多个阶段（stage）。各个阶段都有监督训练，避免过深网络难以优化的问题。
- 使用同一个网络，同时在多个尺度处理输入的特征和响应

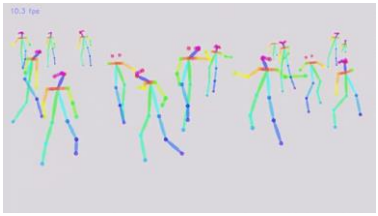


CNN结构演化—检测任务的再次升级，姿态检测

OpenPose SE WEI'CVPR16

- 用各部件响应图来表达各部件之间的空间约束
- 网络分为多个阶段（stage）。各个阶段都有监督训练，避免过深网络难以优化的问题。
- 使用同一个网络，同时在多个尺度处理输入的特征和响应



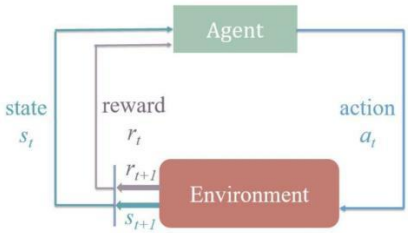


https://posetrack.net/

CNN结构演化—感知决策，规划控制，强化学习

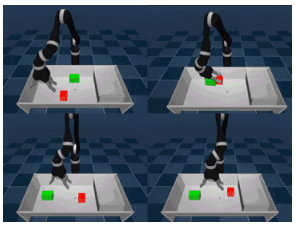
强化学习（Reinforcement Learning)

- 通过 智能体(agent)在环境中不断试错，自我学习纠正
- 主要包含四个元素：agent，环境状态，行动，奖励
- 当agent做出与目标一致的行动时，给出奖励，反之则不给奖励。通过不断训练使得agent做出正确的决策





AlphaGo Zero:从零开始的自我强化学习



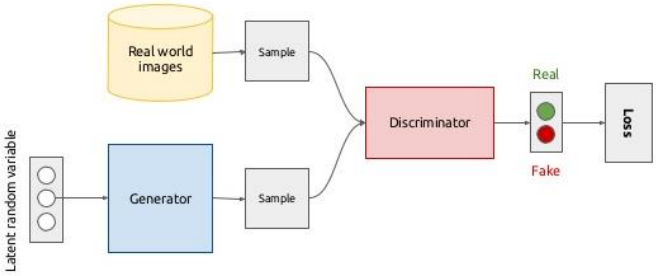
物体抓取控制（SAC-X）

M Riedmiller,Learning by Playing - Solving Sparse Reward Tasks from Scratch,2018

CNN结构演化—从判别到生成，生成对抗网络

GAN(generative adversarial network) Ian'NIPS14

- 生成网络：生成尽可能真实的样本
- 判别网络：对图片的真实程度进行判别。
- 生成网络和判别网络相互博弈，在迭代学习的过程中逐渐进步



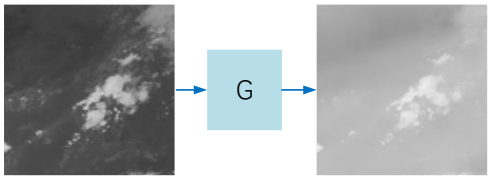
CNN结构演化—从判别到生成，生成对抗网络

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- 生成网络和判别网络相互博弈，在迭代学习的过程中逐渐进步



真实高分辨率道路场景生成(Wang'CVPR18)



输入图像　　输出图像

异谱段红外图像转换

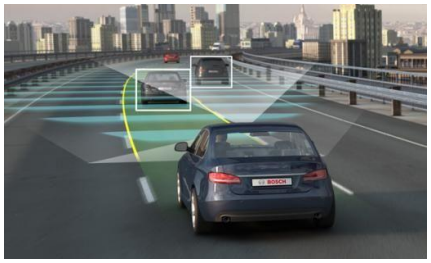
深度学习技术应用

❖ **视觉应用（检测、识别、跟踪、理解、…）**





无人系统应用










智能交互——脑机接口

- 埃隆•马斯克（Elon Musk）脑机接口公司Neuralink，Mindpong（意念乒乓）

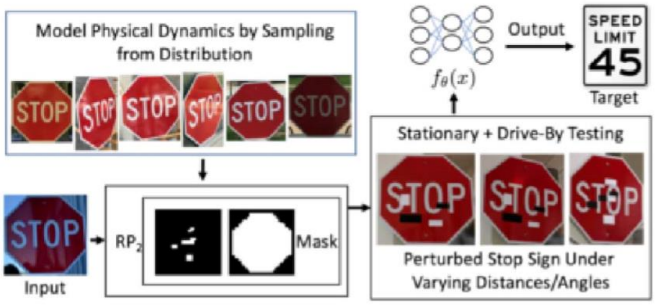


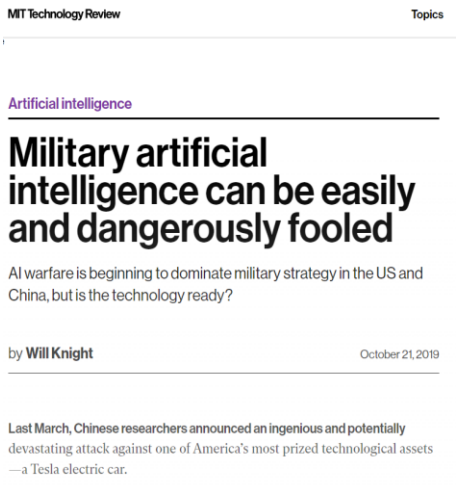



猕猴pager意念控制玩游戏

AI识别对抗

- 对抗样本、生成对抗网络等被统称为**对抗机器学习**技术，已经成为近年来热门的研究方向——反识别，提升模型“鲁棒性”
- 安防监控、无人驾驶、银行，军事人工智能





MIT Technology Review Topics

Artificial intelligence

Military artificial intelligence can be easily and dangerously fooled

AI warfare is beginning to dominate military strategy in the US and China, but is the technology ready?

by **Will Knight** October 21, 2019

Last March, Chinese researchers announced an ingenious and potentially devastating attack against one of America's most prized technological assets —a Tesla electric car.

Etimov等人展示了对抗攻击对现实物理条件有一定**鲁棒性**

第十一讲 卷积神经网络(*Convolutional Neural Networks*)

11.1 背景(Background)

CNN在计算机视觉影响力的一段往事

11.2 卷积核的知识(About Convolutional Kernels)

滤波器组、纹理基元、视觉词典

11.3 卷积神经网络(Convolutional Neural Networks)

卷积层、池化层、全连接层。。。

11.4 应用示例(Case Study)

图像识别应用